Predicting IPR Patent Cancellation at Petition Filing Time

Machine Learning · May 2026

1. Problem Definition

An Inter Partes Review *IPR* is a USPTO administrative procedure used to challenge the validity of an issued patent, a faster and cheaper alternative to district-court litigation. Each case typically resolves after 12–24 months in one of three ways: a Final Written Decision *FWD* cancelling all challenged claims, an FWD where some claims survive, or a non-merits termination (for example settlement, institution denial, or discretionary denial under Fintiv).

Decision supported. At filing time, both sides face a binary call: proceed, settle, or wait. Today, this judgment is made by attorneys reading the petition cold, slow and expensive. A model-produced probability of cancellation, even if only modestly informative, supports this triage when reported alongside its calibration profile.

Target variable. A binary cancelled in {0,1} label where 1 means the FWD held *all* challenged claims unpatentable (or the patent owner conceded by adverse judgment); 0 means any claim survived.
Why supervised learning? Each closed IPR has a public outcome (cancelled or not) recorded by the (USPTO Open Data Portal). The portal also publishes rich structured data per filing.

2. Dataset

We assemble a row-per-trial dataset from the USPTO Open Data Portal by joining four structured surfaces (proceedings, decisions, petitions, patent-application records) plus two PDF surfaces (the petition text, and the FWD text used purely for label construction since the verdict is not exposed in any structured field).

The raw cohort spans 15,030 IPR trials filed between 2012-09-16 and 2026-01-05. After dropping rows with unrecoverable petition text (<5,000 characters) or unfetched patent file wrappers, 14,822 trials remain.

We then apply a 600-day mature-label filter (dropping rows whose petition is more recent than ~20 months ago), since FWDs take a median of ~549 days to issue. Without this, recent trials are mechanically dominated by fast-resolution outcomes (settlements, institution denials), all coded 0, due to survivorship bias: cases close to the present can only either be unresolved or resolved early through non-merits termination. The final modelling cohort is **13,596 trials**, with class balance shifting from 0.1748 to 0.1871, confirmation that the filter was not redundant.

3. Feature Engineering

The single inviolable constraint is T_0 leakage discipline: every feature must be observable strictly before the petition-filing date T_0. The transform layer produces 35 raw features in seven families, summarized below.

Family	n	Examples
Calendar	3	filing_year, filing_month, art_unit_group
Missingness pairs	4	prosecution_span_days(_missing), days_grant_to_petition(_missing)
Patent file-wrapper aggregates	11	n_events, n_assignments, n_distinct_assignees, n_parent_applications, n_pe / ex / aa / ad / iss / maint /...
Assignment recency / CPC breadth	4	days_since_last_assignment, no_recorded_assignment, n_cpc_codes, n_cpc_subclasses
Categorical — OHE	5	technology_center, cpc_section, ptab_era, entity_size, inventor_geo
Categorical — frequency	2	petitioner_real_party, owner_real_party
Petition-text	6	n_grounds, n_grounds, has_sotera_stipulation, mentions_fintiv_factors, petition_text_length

NaN-as-signal. Each nullable numeric feature gets a paired *<name>_missing* indicator before median inputation, so the missingness regime survives the fill. This matters for *prosecution_span_days* and *days_grant_to_petition*: their missing values correlate with patent type (continuations have shorter spans; missing grant dates flag re-issues) and collapsing them into a median would erase the signal.

Rolling priors gated by label-resolution date, not filing date. Six prior-rate features (over petitioner, owner, petitioner×owner pair, technology center, patent number, and PTAB era) plus two open-vocabulary frequency encodings are computed once over the full corpus, with leakage prevented by a strict < gate on label-resolution date: a corpus row contributes to a target row's prior only if its outcome was observable strictly before the target row's T_0.

4. Modelling Pipeline

Preprocessor. The 35 raw features go through two preprocessing stages before entering the model. Firstly, *attach_rolling_encodings* computes leakage-safe categorical encodings once over the full dataset: a *PriorEncoder* applied across six group keys creates historical statistics based on cases resolved before T_0. For every case, it looks at six types of groups/regime keys: petitioner, patent owner, petitioner×owner pair (combination), technology center (area), patent number, and PTAB era (political regime). At the same time, the *FrequencyEncoder* records the number of times the companies' name has been mentioned in the past cases pre-T_0 (unseen companies → 0). Both encoders are strictly time aware. Only using information available before the filing date T_0 of each row, this is implemented through a strict *searchsorted* lookup. Ensuring a row never sees information filed during or after its T_0, whilst still allowing each row to use the full pre- T_0 historical data rather than only its CV fold's training prefix. Secondly, the transformed features go through a per-fold *ColumnTransformer* with two branches: a *OneHotEncoder* for the closed categoricals (with the addition of a *__missing__* category so these values become an explicit one-hot) and a *SimpleImputer(strategy="median")* over the numerical columns, including the raw numerical features and the precomputed rolling encodings. After all these modifications, the design matrix has exactly 82 columns.

The *PriorEncoder* is the most relevant (as per the feature importances) feature engineering component. For each row at T_0 and each group key (e.g., the petitioner), the encoder will compute a: rolling cancellation rate (for rows before T_0) and the number of prior resolved cases. The ordering by *label_resolution_date* rather than by *petition_filing_date* is done on purpose as a co-filed trial whose label is not yet observable at T_0 cannot contribute to the prior.

For unseen groups (a first-time petitioner) the encoder uses the rolling global rate at T_0. This avoids returning zero (which would imply the petitioner has lost every prior case) or the corpus mean (which would incorporate future information and introduce leakage).

Estimator and hyperparameters. The chosen model was selected empirically. We grid-searched both *RandomForestClassifier* and *LogisticRegression* (with *StandardScaler*, saga and elastic-net) over the same time-based forward-walking CV. RandomForest outperformed on the held-out tail (ROC-AUC ≈ 0.61 vs ≈ 0.52 for LR's). Since the LR only performed slightly above pure randomness, we selected RF as the final model.

The selected RF hyperparameters were *n_estimators=500*, *max_depth=None*, *max_features=0.3*, *min_samples_leaf=5*, *class_weight='balanced', random_state=42*. These proved to give the best results. Additionally, we chose RF because of its simplicity when defining feature importance and easier ability to interpret in comparison to linear coefficients on encoded categorical variables.

Validation strategy. Due to the nature of the problem, we used a forward-walking time-series cross-validation on the pre-cutoff dataset and kept a final unseen test set after 2023-01-01 to simulate a real-world scenario deployment. Random K-fold CV would have caused data leakage as it would have mixed future cancellation rates with past predictions through the *PriorEncoder* hence inflating the AUC.

The CV folds are aligned to date boundaries, so no same-day group are in both the train/test splits. This was vital because joinder cases (companies join IPR) and multi-petition campaigns are often filed in same-day and would be highly correlated. CV scoring includes: ROCAUC, average precision, and F1; whilst the held-out tail is only scored once and never used for model selection or hyperparameter tuning to keep evaluation honest.

Choice of evaluation metrics. The legal user this model is intended for does not want a label, they want probability for an expected-value decision, for this reason we report three dimensions: ranking (ROC-AUC, average precision), calibration (a reliability diagram comparing predicted probabilities vs observed frequencies), and threshold metrics (precision / recall / F1 on the rare positive class). Accuracy metrics are misleading as the held-out cancellation rate is only ~22.7%. A classifier that predicts 0 achieves ~77.3% (not useful).

Notes:

Firstly, we considered post-hoc probability calibration (Platt/sigmoid scaling on top of the RF score), but on such a small split, fitting the calibrator required holding out additional validation training data would have guided our AUC toward random. We therefore chose to show the uncalibrated forest probability with the reliability diagram to communicate the miscalibration to the user.

Second, we chose not to hard-code a classification threshold because misclassification costs are dependent on the context: a false negative makes a potentially winnable trial not to file the petition; a false positive could encourage proceeding with a losing trial. Instead, we output raw probabilities; giving downstream users the freedomto pick thethreshold that matches their risk and legal profile.

5. Results

Metric	Value	Interpretation
Train rows / Held-out rows	11,689 / 1,907	Cutoff 2023-01-01; tail 433 cancelled (22.7%)
CV ROC-AUC (5-fold, mean ± std)	0.551 ± 0.036	Modest but above-chance signal on the pre-cutoff slice
Held-out ROC-AUC	0.611	Real but limited ranking power at T0
Held-out Accuracy	0.776	Barely exceeds the 0.773 majority baseline by ~0.3pp; misleading
Held-out Precision (class 1)	1.000	Of 5 trials flagged 1 at the 0.5 cutoff, ~100% truly cancel
Held-out Recall (class 1)	0.012	Catches only ~1% of true cancellations at default 0.5 cutoff
Held-out F1 (class 1)	0.023	Conservative on the rare class
Calibration	see Fig. 1	Mid-range bins (n=1,358 dominant) mildly over-confident; tail bins thin

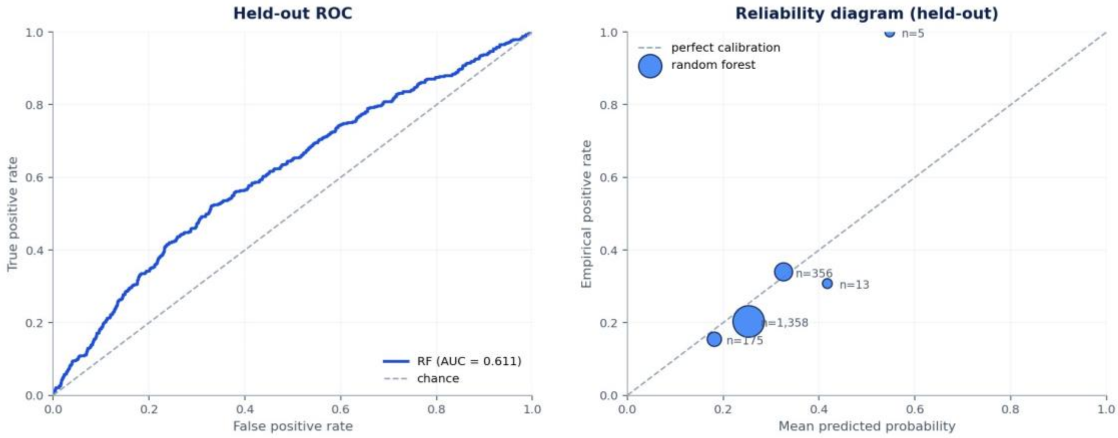

Figure 1. Held-out ROC curve (left) and reliability diagram (right). The ROC curve on the left shows that the model has limited but real ranking ability, with an AUC of 0.611. The reliability diagram on the right shows that the model is somewhat overconfident. In the largest bin, the model predicts an average cancellation probability of 0.253, but the actual cancellation rate is only 0.203.

Three findings emerge from the results. First, the model has a **real but modest ranking signal**. A held-out ROC-AUC of 0.611 means the model performs better than random guessing, but it is not strong enough to be used as a stand-alone predictor. Second, the default 0.5 threshold performs poorly because cancellations are relatively rare. At this cutoff, the model predicts only 5 cancellations, so it achieves perfect precision but captures only about 1% of actual cancellations. Third, because the data is time-based, the dataset cannot be randomly shuffled before training. Doing so would create leakage from future cases into past cases.

6. Critical Reflection

Construct validity (label).

The label **cancelled** measures whether the patent was legally cancelled. However, it does not fully measure whether the petitioner achieved its business objective. For example, a petitioner may obtain a favorable settlement or license even if the patent is not formally cancelled. In that case, the trial is coded as 0, even though the petitioner may still consider the outcome successful. Similarly, a partial final written decision that cancels only some challenged claims is also coded as 0, even though it may still weaken the patent. Therefore, the label is clear in the legal record, but it is an imperfect measure of the real business outcome.

Construct validity (text features).

The text features are based on regular expressions applied to PDF-extracted legal text. This means they may capture **legal writing style** rather than the actual legal strength of the argument. For instance, the model may detect whether lawyers used a standard legal phrase, but that does not necessarily mean the argument is strong, as is the case of the Fintiv or Sotera-related features. Some weak petitions may use the correct wording, while some strong petitions may express the argument differently. Three candidate text features were removed for this reason. The remaining Tier-A features passed a manual audit on the 122-trial integration cohort, but the risk does not disappear completely. A future version could compare these regex-based features with LLM-based evaluations of legal argument quality.

Distributional shift.

The ptab_era feature is structural: a USPTO Director change literally redefines $P(Y \mid X)$ through guidance memos and discretionary-denial policy. The latest era (trump_2_post_vidal, January 2025 onward) is severely under-represented in training, yet 100% of deployment inputs would lie in it. Forward-walking CV partly mitigates by scoring on later folds, but a regime-aware retraining cadence, at minimum on every Director memo, is an operational requirement, not a nice-to-have.

Information ceiling at T_0

The hardest limit is informational rather than methodological. Leakage discipline excludes by construction every signal that accrues after filing, the institution decision, claim-construction order, deposition-stage motions, expert reports; all of which the IPR-outcome literature identifies as the dominant predictors of cancellation. A 0.611 held-out AUC at T_0 is therefore close to a T_0 ceiling rather than evidence of an under-tuned model; meaningful gains likely require either (a) a staged-prediction setup with a refresh after institution, or (b) substantially richer petition-text representations (an LLM-derived legal argument quality score). Both are deferred to future work.

Honest deployment posture.

We treat the model as *decision support*, not autonomous decisioning. What the pipeline actually outputs today is a raw probability score per trial, uncalibrated by design and without an uncertainty band. The reliability diagram (Figure 1, right) is shipped alongside the score so the 0.1 – 0.3 over-confidence is visible to the user rather than hidden. Attorneys, not the model, own the call. The metric we would track in production is not held-out AUC but the drift of the reliability curve on each subsequent year's FWDs, with an automatic retrigger to retrain on regime change.